

Summary

In this project, I analyzed employee attrition data to understand why employees leave the company and to identify employees who may be at a higher risk of leaving. The analysis showed that the three strongest factors predicting employee attrition are **Job Role (Laboratory Technician)**, **Overtime**, and **Frequent Business Travel**. Among all departments, the Sales department, particularly Sales Representatives, had the highest attrition rate. The findings also indicate that salary alone does not explain employee attrition, as work-related factors such as overtime, job role, business travel, and total working years have a much stronger influence on an employee's decision to leave.

Based on these insights, HR should focus on reducing excessive overtime and promoting a healthier work-life balance for employees. In addition, regular retention meetings and career development programs should be conducted for employees in high-risk roles, especially Sales Representatives and Laboratory Technicians. These measures can help improve employee satisfaction and reduce attrition. Although the predictive model provides valuable insights, HR should also consider individual employee circumstances before making any final decisions.